

Research project

in Physics

Static dephasing through field inhomogeneities during off-resonant RF-pulses

Johanna Link

Supervisor: Prof. Dr. Moritz Zaiss

Submission date: January 30, 2026

Abstract

During radiofrequency (RF) excitation in magnetic resonance (MR) experiments, the magnetization evolves under the influence of pulse strength, its off-resonance, and relaxation. The latter is often neglected in practical simulations to reduce computational cost, an approximation that must eventually break down for a certain pulse duration. This project aims to analyze the impact of transverse dephasing due to field inhomogeneities on magnetization throughout the pulse and to test the validity of common approximations for this complex behavior. To this end, several solution approaches to the off-resonant Bloch equations are developed and compared to analyze the influence of T_2' dephasing on the magnetization trajectory. The simulation is then applied to slice selection, where the validity ranges of the hard-pulse and the T_2^* approximation are investigated. The results show that both approximations are accurate only for short RF pulses and break down once the pulse duration becomes comparable to T_2' .

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List of Abbreviations

CEST chemical exchange saturation transfer. 3, 26

MR magnetic resonance. 1, 5, 18, 19, 26

MRI magnetic resonance imaging. 3

NMR nuclear magnetic resonance. 3

RF radiofrequency. 1, 3–9, 11, 12, 14, 15, 17–26

RMS root mean squared. 12, 17

UTE ultrashort echo time. 3

ZTE zero echo time. 3

1 Introduction

Magnetic resonance imaging (MRI) relies on the interaction between nuclear spin systems, externally applied static magnetic fields, and RF pulses [1]. During RF excitation, the magnetization evolves under the influence of the time-dependent RF field, relaxation processes, and resonance offsets that arise from gradients or static field inhomogeneities. While the Bloch equations provide a description of the macroscopic magnetization dynamics [2], many practical simulations use simplifying assumptions to reduce computational effort. A common approximation is the hard-pulse approximation [3], in which the effect of the pulse is modeled as an instantaneous rotation followed by a free-evolution period in which relaxation takes place.

This assumption is used in sequence design, slice-profile simulations, and quantitative modeling [4, 5], although its validity is not guaranteed for all regimes. In particular, long RF pulses or short relaxation times may lead to significant discrepancies between the hard-pulse approximation and the true magnetization dynamics. These conditions are relevant, for example, during the long shaped saturation pulses employed in chemical exchange saturation transfer (CEST) [6], when using adiabatic pulses [7] or in ultrashort echo time (UTE) and zero echo time (ZTE) MRI. In this last case, the short T_2 and T_2^* of the examined tissues are often comparable to the pulse duration in spite of the short pulses [8].

The goal of this research project is to systematically investigate how relaxation during off-resonant RF pulses affects the magnetization trajectory and how strongly the hard-pulse approximation breaks down across realistic parameter ranges. To this end, different solution approaches to the full Bloch equations are developed and verified against each other, and are then ensemble-averaged over static resonance offsets in a Cauchy distribution. These models are employed to analyze the signal evolution during an off-resonant pulse and are finally applied to slice selection to determine the practical limit of the hard-pulse and T_2^* approximation.

2 Theoretical background

2.1 Physical context

In the following section, the principles of nuclear magnetic resonance (NMR) will be explained in the context of relaxation and off-resonance, based on Levitt [1]. In the Bloch picture, NMR is described in terms of a macroscopic magnetization vector $\vec{M}$. In thermal equilibrium, $\vec{M}$ is aligned with the static field $\vec{B}_0$. This is usually defined as the z-direction, so $\vec{M}_0 = M_0 \hat{z}$. An applied RF field at or near the resonance (Larmor) frequency rotates $\vec{M}$ away from equilibrium, and its subsequent evolution is governed by RF pulses, relaxation, and resonance offsets.

Relaxation After excitation, the magnetization vector does not remain in the transverse plane indefinitely. Interactions with the surrounding lattice and with other spins drive the system back toward thermal equilibrium. There are two unrecoverable relaxation processes to be considered: longitudinal relaxation, characterized by the time constant T_1 , describes the recovery of the z-component of the magnetization. Transverse relaxation, described by T_2 , reflects the loss of phase coherence among spins in the transverse plane caused by spin-spin interaction, leading to an effective shortening of the magnetization vector.

In the on-resonant case, and in the absence of RF excitation, these relaxation mechanisms can be described as exponential decays of the magnetization components,

$$M_z(t) = M_0 - (M_0 - M_z(0))e^{-t/T_1}, \quad M_{xy}(t) = M_{xy}(0)e^{-t/T_2}. \quad (1)$$

In practice, the effectively measured transverse relaxation time is shorter than the intrinsic T_2 , as static field inhomogeneities cause additional dephasing [9]. In the free-evolution case, this reversible dephasing can be recovered by applying a 180° refocusing pulse. The pulse inverts the transverse magnetization vectors in the rotating frame, such that spins that were precessing faster are placed behind the slower ones. During subsequent free evolution, this relative phase dispersion is unwound, leading to a rephasing of the magnetization at the echo time. It is then also possible to approximate this static dephasing as an exponential, characterized by the time constant T'_2 . The combined transverse decay is then described by T_2^* , defined as

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T'_2} \quad (2)$$

This treatment, in which static dephasing is incorporated into an effective transverse relaxation time, is referred to as the T_2^* approximation in this report.

Off-resonance Apart from relaxation, additional phase dispersion arises whenever spins experience different resonance frequencies. One example is slice selection, where spins experience a deterministic, position-dependent frequency offset due to the applied slice selection gradient. In this report, this offset is denoted by $\Delta\omega_0$ and serves as the central off-resonance around which additional microscopic frequency variations - accounted for phenomenologically by T'_2 - are distributed. In slice selection, the position along one axis is made frequency-dependent by applying a field gradient that imposes a resonance offset $\Delta\omega_0 = \gamma Gz$. Here, the gradient G is applied along the z-axis, and z denotes the spatial position of the magnetization vector $\vec{M}(z)$ under consideration. The finite bandwidth of a slice-selective RF pulse means that only spins with a local Larmor frequency lying near the pulse center frequency are on-resonant, while all others experience varying degrees of off-resonance. Because RF excitation acts simultaneously on a continuum of resonance offsets, the observed magnetization must be obtained as an ensemble average over the corresponding $\Delta\omega_0$ values within the slice.

In simulations of the MR response, relaxation during the pulse is often neglected because T_1 and T_2^* are several orders of magnitude longer than the typical millisecond-scale RF duration [8]. The RF pulse is therefore modeled as a relaxation-free, instantaneous rotation followed by free relaxation. However, this approximation must eventually fail for sufficiently long pulses or large relaxation rates, making it necessary to account for relaxation during excitation. The following sections introduce how the dependence of the MR signal on resonance offsets is modeled and how the ensemble average is obtained.

2.2 The Bloch equations in the off-resonant case

Bloch equations and effective field The dynamics of the nuclear magnetization vector $\vec{M} = (M_x, M_y, M_z)$ in the absence of gradients during RF excitation are governed by the Bloch equations derived by Bloch [2]. In the rotating frame, defined as the frame rotating at the RF carrier frequency ω_{rf} , they are

$$\frac{d\vec{M}}{dt} = \gamma \vec{M} \times \vec{B}_{\text{eff}} - \begin{pmatrix} M_x/T_2 \\ M_y/T_2 \\ (M_z - M_0)/T_1 \end{pmatrix} \quad (3)$$

where T_1 and T_2 are the longitudinal and transverse relaxation times, γ is the gyromagnetic ratio, and M_0 is the thermal equilibrium magnetization. The effective magnetic field is

$$\vec{B}_{\text{eff}}(t) = (B_{1x}(t), B_{1y}(t), \Delta B_0), \quad (4)$$

where ΔB_0 is the static field inhomogeneity, and $\Delta\omega_0 = \gamma \Delta B_0$ is the corresponding resonance offset.

It is often convenient to work with the effective precession vector

$$\vec{\omega}_{\text{eff}}(t) = \gamma \vec{B}_{\text{eff}}(t) = (\gamma B_{1x}(t), \gamma B_{1y}(t), \Delta\omega_0), \quad (5)$$

so that Bloch dynamics are described purely as rotations about $\vec{\omega}_{\text{eff}}(t)$, modified by relaxation.

Matrix form of the Bloch equations Writing equation (3) in matrix form and choosing the coordinate system such that the y-component of the RF nutation frequency is 0, the Bloch equations take the form

$$\dot{\vec{M}} = \mathbf{A} \vec{M} + \vec{C}, \quad (6)$$

with

$$\mathbf{A} = \begin{pmatrix} -R_2 & +\Delta\omega & 0 \\ -\Delta\omega & -R_2 & +\omega_1 \\ 0 & -\omega_1 & -R_1 \end{pmatrix}, \quad \vec{C} = \begin{pmatrix} 0 \\ 0 \\ M_0 R_1 \end{pmatrix}. \quad (7)$$

Here

$$R_1 = 1/T_1, \quad R_2 = 1/T_2, \quad \omega_1 = \gamma B_1, \quad \Delta\omega = \omega_0 - \omega_{\text{rf}},$$

denote the longitudinal and transverse relaxation rates, the RF nutation frequency, and the resonance offset in the rotating frame.

In the following, an analytic solution of the Bloch equations is derived for the time evolution of the magnetization over a finite interval t . This treatment assumes that all system parameters, including relaxation rates, off-resonance, and the RF amplitude ω_1 , remain constant over the considered time interval.

Shift to the homogeneous system The steady state magnetization satisfies $\dot{\vec{M}} = 0$; hence,

$$\vec{M}_\infty = -\mathbf{A}^{-1}\vec{C}. \quad (8)$$

Introducing the shifted variable $\vec{m}(t) = \vec{M}(t) - \vec{M}_\infty$, equation (6) reduces to the homogeneous system

$$\dot{\vec{m}}(t) = \mathbf{A}\vec{m}(t), \quad (9)$$

with the solution

$$\vec{m}(t) = e^{\mathbf{A}t} \vec{m}(0). \quad (10)$$

This transformation removes the inhomogeneous driving term and allows the dynamics to be expressed solely in terms of the system matrix $\mathbf{A}$. The resulting time evolution is therefore fully determined by the propagator of this system, which is given by the matrix exponential $e^{\mathbf{A}t}$. To obtain an explicit, closed-form expression for this propagator, the system matrix $\mathbf{A}$ must be diagonalized.

Diagonalization If $\mathbf{A}$ is diagonalizable, one may write $\mathbf{A} = \mathbf{D}\mathbf{\Lambda}\mathbf{D}^{-1}$, where the columns of $\mathbf{D}$ are the eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$ contains the eigenvalues. Then

$$e^{\mathbf{A}t} = \mathbf{D} e^{\mathbf{\Lambda}t} \mathbf{D}^{-1}. \quad (11)$$

Undoing the steady-state shift results in the formal solution

$$\vec{M}(t) = \mathbf{D} e^{\mathbf{\Lambda}t} \mathbf{D}^{-1} (\vec{M}(0) + \mathbf{A}^{-1}\vec{C}) - \mathbf{A}^{-1}\vec{C}. \quad (12)$$

This expression forms the basis for the analytic and numerical treatment in the following section. The vector $\mathbf{A}^{-1}\vec{C}$ appearing in equation (12) is the steady-state magnetization for a constant off-

resonant RF field. Evaluating the inverse explicitly gives

$$\mathbf{A}^{-1}\vec{C} = \begin{pmatrix} \frac{M_0 R_1 \Delta\omega \omega_1}{R_2 \omega_1^2 + R_1 (\Delta\omega^2 + R_2^2)} \\ \frac{M_0 R_1 R_2 \omega_1}{R_2 \omega_1^2 + R_1 (\Delta\omega^2 + R_2^2)} \\ \frac{M_0 R_1 (\Delta\omega^2 + R_2^2)}{R_2 \omega_1^2 + R_1 (\Delta\omega^2 + R_2^2)} \end{pmatrix}. \quad (13)$$

2.3 Modelling static field inhomogeneity

In the presence of static field inhomogeneity, different spins experience different resonance offsets $\Delta\omega$. The observed magnetization is the ensemble average of the single isochromat magnetization $\vec{M}(t, \Delta\omega)$ from equation (12) [10]. An isochromat represents a subset of spins with identical resonance frequency that evolve coherently under the applied RF and static fields. Here, the RF amplitudes ω_1 and relaxation rates R_1, R_2 are treated as constant, so that each isochromat's magnetization depends only on time t and its resonance offset $\Delta\omega$. The total observed magnetization is then obtained by averaging over the distribution of offsets:

$$\vec{M}(t) = \int_{-\infty}^{\infty} \vec{M}(t, \Delta\omega) p(\Delta\omega) d\Delta\omega. \quad (14)$$

A widely used model for frequency dispersion is the Lorentzian lineshape [11], which results from a Cauchy distribution [12],

$$p(\Delta\omega) = \frac{1}{\pi} \frac{R_2'}{(\Delta\omega - \omega_0)^2 + R_2'^2}, \quad R_2' = 1/T_2'. \quad (15)$$

This corresponds to a static distribution of off-resonances which, under the assumption of a static Cauchy-distributed ensemble of off-resonances, results in an exponential T_2' decay in the free evolution case [9].

2.4 Numerical quadrature

The remaining integral is solved using Gauss-Legendre quadrature [13].

Applied to the present problem, the integral over $\Delta\omega \in (-\infty, \infty)$ is first mapped onto a finite domain. Using the substitution

$$\Delta\omega = \Delta\omega_0 + r_2' \tan \theta, \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad (16)$$

the ensemble average becomes

$$\overline{\vec{M}}(t) = \frac{1}{\pi} \int_{-\pi/2}^{+\pi/2} \vec{M}(t, \Delta\omega_0 + r'_2 \tan \theta) d\theta. \quad (17)$$

The remaining integral over the compact interval $[-\pi/2, \pi/2]$ can then be evaluated by linearly mapping the Gauss-Legendre nodes $x_k \in [-1, 1]$ to θ_k ,

$$\theta_k = \frac{\pi}{2} x_k, \quad w'_k = \frac{\pi}{2} w_k, \quad (18)$$

yielding the numerical approximation

$$\overline{\vec{M}}(t) \approx \frac{1}{\pi} \sum_{k=1}^n w'_k \vec{M}(t, r'_2 \tan \theta_k). \quad (19)$$

2.5 Cayley-Klein formulation

An alternative approach to solving the Bloch equations in the rotating frame via matrix exponentials is the Cayley-Klein (Spinor) formalism described by Le Roux, Nishimura, and Macovski [3]. In this representation, each infinitesimal rotation of the magnetization about the instantaneous effective field $\vec{\omega}_{\text{eff}} = \gamma \vec{B}_{\text{eff}} = (\gamma B_{1x}, \gamma B_{1y}, \Delta\omega_0)$ is expressed as a 2×2 complex matrix acting on a spinor. This approach is algebraically compact and particularly convenient for simulating RF pulses because a time-dependent field is treated as a sequence of small rotations whose matrices can be multiplied efficiently. As the formalism can only be applied to unitary matrices, relaxation must be neglected.

Each spinor matrix representing a small rotation has the form

$$Q = \begin{pmatrix} a & -b^* \\ b & a^* \end{pmatrix} \quad (20)$$

with the Cayley-Klein parameters a and b , for which $|a|^2 + |b|^2 = 1$ is valid. For an effective field of magnitude $|\omega_{\text{eff}}| = \sqrt{|\omega_1|^2 + \Delta\omega^2}$ and in the direction $\hat{n} = (\omega_{1x}, \omega_{1y}, \Delta\omega)/|\omega_{\text{eff}}|$, the infinitesimal rotation has the parameters

$$a = \cos\left(\frac{|\omega_{\text{eff}}|\Delta t}{2}\right) - in_z \sin\left(\frac{|\omega_{\text{eff}}|\Delta t}{2}\right) \quad (21)$$

and

$$b = (-in_x + n_y) \sin\left(\frac{|\omega_{\text{eff}}|\Delta t}{2}\right). \quad (22)$$

After a constant amplitude RF pulse and for the initial equilibrium $(M_x, M_y, M_z) = (0, 0, 1)$, the final

magnetization is given by

$$M_{xy} = M_x + iM_y = 2a * b \quad (23)$$

$$M_z = |a|^2 - |b|^2. \quad (24)$$

For a time-dependent RF pulse, the total spinor is obtained by chaining infinitesimal rotations at each time step, concatenated as $Q_{tot} = Q_N Q_{N-1} \dots Q_1$. This yields the transverse and longitudinal magnetization for each resonance offset $\Delta\omega$.

3 Solution approaches

3.1 Analytic solution

The formal solution in equation (12) fully determines the time evolution during a constant RF pulse once the eigenvalues and eigenvectors of $\mathbf{A}$ are known. In the next section, this diagonal form is evaluated explicitly following Johnston [14] for the off-resonant Bloch matrix in an attempt to find a closed analytic expression for $\vec{M}(t)$.

Eigenvalues and effective field representation The characteristic polynomial of the Bloch matrix,

$$\det(\mathbf{A} - \lambda \mathbf{1}) = 0, \quad (25)$$

is cubic in λ . As explained in subsection A.1, the polynomial admits the factorization

$$(\lambda + R_{1\rho})[(\lambda + R_{2\rho})^2 + \Omega^2] = 0,$$

with the eigenvalues

$$\lambda_1 = -R_{1\rho}, \quad (26)$$

$$\lambda_{2,3} = -R_{2\rho} \pm i\Omega. \quad (27)$$

Here, $\Omega = \sqrt{\Delta\omega^2 + \omega_1^2}$ is the magnitude of the effective field and the corresponding tilt angle

$$\cos \theta = \frac{\Delta\omega}{\Omega}, \quad \sin \theta = \frac{\omega_1}{\Omega}, \quad (28)$$

allows the relaxation rates along and perpendicular to the effective field to be expressed in a compact form

$$R_{1\rho} = R_1 \cos^2 \theta + R_2 \sin^2 \theta, \quad (29)$$

$$R_{2\rho} = R_2 \cos^2 \theta + R_1 \sin^2 \theta. \quad (30)$$

These three eigenmodes describe slow relaxation along the effective field and two damped precessional modes that are transverse to it.

Eigenvectors and diagonalization The eigenvectors follow from the homogeneous system

$$(\mathbf{A} - \lambda_k \mathbf{1}) \vec{v}_k = 0. \quad (31)$$

Closed expressions obtained via cofactors are provided in Appendix A.3, but they are lengthy and not required for this discussion.

To obtain a compact analytic form of the propagator $e^{\mathbf{A}t} = \mathbf{D}e^{\mathbf{\Lambda}t}\mathbf{D}^{-1}$, it is convenient to work with the approximation $R_1 \approx R_2$, which considerably simplifies the structure of the eigenvectors. In this case, the matrix of eigenvectors can be written as

$$\mathbf{D} = \begin{pmatrix} \sin \theta & \cos \theta & \cos \theta \\ 0 & i & -i \\ \cos \theta & -\sin \theta & -\sin \theta \end{pmatrix}, \quad \mathbf{D}^{-1} = \begin{pmatrix} \sin \theta & 0 & \cos \theta \\ \frac{\cos \theta}{2} & -\frac{i}{2} & -\frac{\sin \theta}{2} \\ \frac{\cos \theta}{2} & +\frac{i}{2} & -\frac{\sin \theta}{2} \end{pmatrix}. \quad (32)$$

Homogeneous propagator To make the diagonalized propagator immediately accessible in a compact form, the following abbreviations are used:

$$E_1(t) = e^{-R_1 \rho t}, \quad E_2(t) = e^{-R_2 \rho t}, \quad c = \cos(\Omega t), \quad s = \sin(\Omega t),$$

With those, the homogeneous propagator $\mathbf{X}(t) = \mathbf{D}e^{\mathbf{\Lambda}t}\mathbf{D}^{-1}$ can be written as

$$X_{11} = \sin^2 \theta E_1 + \cos^2 \theta E_2 c, \quad X_{12} = \cos \theta E_2 s, \quad X_{13} = \sin \theta \cos \theta (E_1 - E_2 c), \quad (33)$$

$$X_{21} = -\cos \theta E_2 s, \quad X_{22} = E_2 c, \quad X_{23} = \sin \theta E_2 s, \quad (34)$$

$$X_{31} = \sin \theta \cos \theta (E_1 - E_2 c), \quad X_{32} = -\sin \theta E_2 s, \quad X_{33} = \cos^2 \theta E_1 + \sin^2 \theta E_2 c. \quad (35)$$

The magnetization of a single isochromat $\Delta\omega$ then follows from the formal solution in equation (12).

Cauchy-distributed ensemble In the relaxation-free limit $R_1 = R_2 = 0$, the single-isochromat solution reduces to a rotation about the effective field. For an initial equilibrium magnetization

$\vec{M}(0) = M_0 \hat{z}$, the components are

$$M_x(t, \Delta\omega) = \frac{\Delta\omega \omega_1}{\Delta\omega^2 + \omega_1^2} \left[1 - \cos(\sqrt{\Delta\omega^2 + \omega_1^2} t) \right] M_0, \quad (36)$$

$$M_y(t, \Delta\omega) = \frac{\omega_1}{\sqrt{\Delta\omega^2 + \omega_1^2}} \sin(\sqrt{\Delta\omega^2 + \omega_1^2} t) M_0, \quad (37)$$

$$M_z(t, \Delta\omega) = \left[\frac{\Delta\omega^2}{\Delta\omega^2 + \omega_1^2} + \frac{\omega_1^2}{\Delta\omega^2 + \omega_1^2} \cos(\sqrt{\Delta\omega^2 + \omega_1^2} t) \right] M_0. \quad (38)$$

The ensemble-averaged components are then obtained by performing the weighted integral given in subsection 2.3. However, even with the approximation $R_1 = R_2 = 0$, these integrals do not admit closed-form solutions and must therefore be evaluated numerically.

3.2 Numerical Solution

Numerical reference solution of the matrix exponential To test the validity of the approximation $R_1 \approx R_2$ made in the previous calculation, a numerical reference solution was obtained by directly exponentiating the matrix $\mathbf{A}$ and assembling it with the driving term $\vec{C}$ according to equation (12). The matrix exponential $e^{\mathbf{A}t}$ is evaluated directly using `scipy.linalg.expm`. For each time point t_i , the magnetization is computed as

$$\vec{M}_i(t) = e^{\mathbf{A}t} (\vec{M}(0) - \vec{M}_\infty) + \vec{M}_\infty \quad (39)$$

with the steady state of the driven system $M_\infty = \mathbf{A}^{-1} \vec{C}$ calculated once in the beginning. This approach computes the exact solution of the linear Bloch system for a given set of input parameters up to numerical precision.

Comparison of analytic and numerical approach for one isochromat Figure 1a compares the analytic solution to the exact numerical one for $T_2 = 0.5 T_1$ and a starting magnetization $M_0 = (0, 0, 1)^\top$ under a constant RF pulse of $\omega_1 = 2\pi$ rad/s. As the approximation $R_2 \approx R_1$ is no longer valid in that regime, the analytic solution deviates slightly from the numerical one. As figure 1b shows, the analytical solution is exact for $R_1 = R_2$. In the chosen case with $\Delta\omega = 2\pi$ rad/s and a constant RF pulse with $\omega_1 = 2\pi$ rad/s, the error is acceptable for $R_2 > 0.2 T_1$ and decreases as R_2 approaches R_1 .

3.3 Integration through Gauss-Legendre Quadrature

Even with the simplifications $M_0 = (0, 0, 1)^\top$ and $R_1 = R_2 = 0$, the ensemble integral does not have an analytic closed form; therefore, the integral must be performed numerically. This was im-

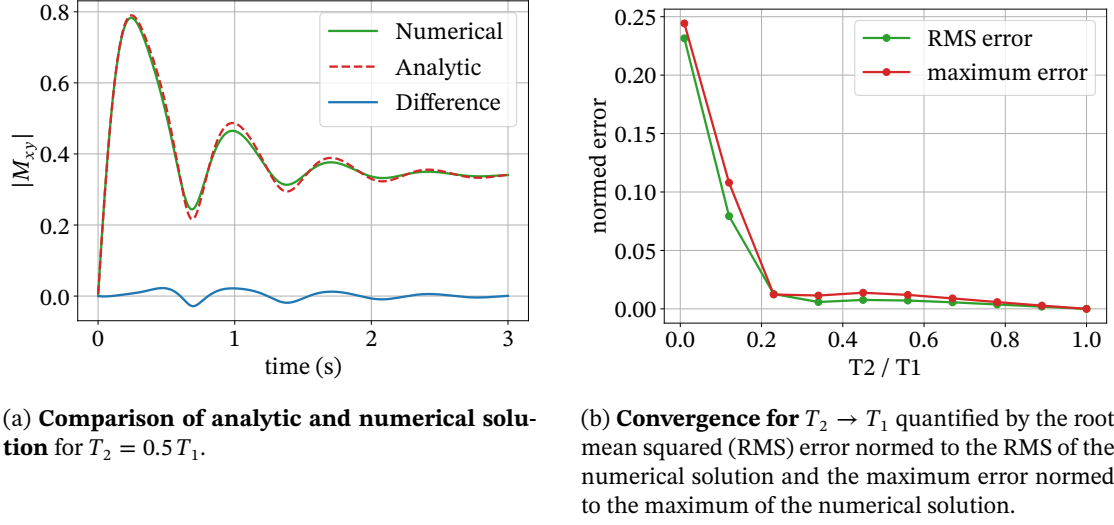


Figure 1: Comparison of numerical and analytic magnetization evolution for one isochromat.

plemented using the Gauss-Legendre quadrature described in subsection 2.4.

The quadrature points and their weights are calculated by the `polynomial.legendre.legauss` function provided by `numpy`. For each Gauss-Legendre point, the Bloch equations are then solved by the chosen function: either the analytic approximation or the numerical solution of a single isochromat. These magnetizations are then weighted by the quadrature weight and normalized by π , the normalization constant of the Lorentzian.

4 Simulation

The analytic solution is efficient for constant ω_1 , but more realistic, time-modulated RF pulses require sequential evaluation and do not allow parallelization. To address this limitation, a simulation by Prof. Dr. Tony Stöcker published as an exercise in the `pulseq` tutorial `MR-Physics-with-Pulseq` [15] based on the Cayley-Klein formulation explained in subsection 2.5 was expanded to include relaxation. Accelerated with Numba, this approach provides a computationally efficient alternative. The implementation is provided in a public repository [16].

The original simulation computes the excitation profile from the full Bloch rotation using Cayley-Klein spinor parameters (see subsection 2.5) and compares the resulting transverse magnetization to that obtained under the small-tip-angle approximation, where the transverse magnetization is assumed to be proportional to the integral of the RF pulse, so that the excitation profile can be obtained by Fourier transforming the pulse envelope.

To model the time-dependent behavior of the magnetization under RF excitation, off-resonance, and relaxation, this Bloch simulation was extended to include T_1 and T_2 relaxation and to incorporate T_2'

by averaging over an ensemble of off-resonant isochromats.

Off-resonance distribution To model the transverse decay caused by static field inhomogeneities, the trajectories of a number of isochromats with specific detunings $\Delta\omega$ are computed. The detunings are assumed to be Cauchy-distributed around the off-resonance $\Delta\omega_0$, which is, for example, caused by slice-selection gradients. The ensemble averaging over the off-resonance offsets $\Delta\omega$ then corresponds to static field inhomogeneity and produces an exponential T_2' decay in the free-evolution limit.

Time stepping Each time-integration loop follows the Cayley-Klein simulation: first, the effective field B_{eff} is built, and the rotation angle is computed $\phi = -|B_{\text{eff}}| \Delta t$. The effective field is then normalized to obtain the rotation axis. This is repeated at every time step across multiple off-resonances $\Delta\omega$.

T_1 and T_2 Relaxation After each rotation, relaxation is applied analytically according to

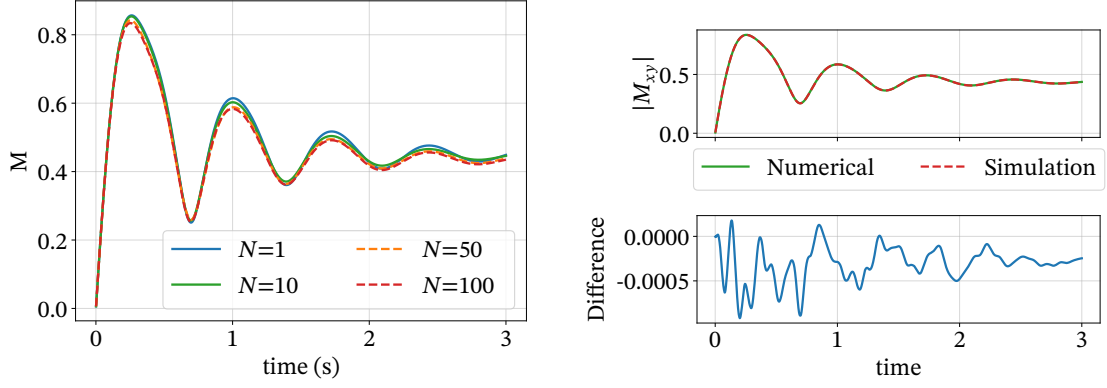
$$M_{xy}(t + \Delta t) = M_{xy}(t)e^{-\Delta t/T_2}, \quad M_z(t + \Delta t) = 1 - (1 - M_z(t))e^{-\Delta t/T_1}. \quad (40)$$

Figure 2a shows the simulation for different numbers of isochromats N for constant $\omega_1 = 2\pi \text{ rad s}^{-1}$ and $\Delta\omega_0 = 2\pi \text{ rad s}^{-1}$, and relaxation times $T_1 = 1 \text{ s}$, $T_2 = 0.8 \text{ s}$, and $T_2^* = 0.7 \text{ s}$. For the chosen parameters, the change is negligible after $N = 50$ isochromats. This is supported by figure 2b, where the simulation for $N = 50$ isochromats agrees well with the solution obtained through quadrature ensemble-averaging over the single-isochromat solution, which was calculated by directly solving the matrix equations numerically.

4.1 Computation times

As table 1 shows, the numba-accelerated Cayley-Klein simulation is the most efficient approach once the function has been executed for the first time. When only a few computations at constant ω_1 are needed, the expense of the first computation makes the analytic approximation the better option. However, the vectorization employed to accelerate computation cannot be applied to the time-dependent ω_1 case. The full numerical approach takes significantly longer but yields accurate results for all combinations of T_1 and T_2 .

Concerning the ensemble average, computing each $\Delta\omega$ sequentially in a `for` loop is straightforward but inefficient. The quadrature selects evaluation points more strategically, requiring fewer points to achieve the same accuracy, and is therefore more efficient. This efficiency can be increased by precompiling the quadrature with numba.



(a) **Increasing number of off-resonances** contributing to the ensemble average of the Cayley-Klein simulation.

(b) **Comparison of the simulation to the numerical solution** for $N = 50$ isochromats in the simulation and 100 Gauss-Legendre points in the numerical integration.

Figure 2: Comparison of numerical and simulated magnetization evolution, integrated to include T_2' effects.

Table 1: Execution time comparison for the single-isochromat solution approaches for $n_t = 2000$ time steps.

Method	Time per call (ms)
Numeric solution	1.547
Analytic approximation	0.225
Cayley-Klein (after first computation)	0.050
Cayley-Klein (warm-up)	896.740

5 Analysis of T_2' behavior

5.1 Convergence for long time spans

In the following, the long-time behavior of the T_2' decay is analyzed in two cases. First, the general case including longitudinal and transverse relaxation is considered, in which each isochromat reaches a steady state under continuous RF driving. This is then contrasted with the relaxation-free limit $T_1, T_2 \rightarrow \infty$, which isolates the effect of static frequency dispersion alone.

Behavior in the presence of T_1 and T_2 relaxation In this regime, the transverse magnetization of each isochromat converges to a time-independent steady-state value that depends on its resonance offset. The experimentally observed signal is obtained by averaging these steady-state solutions over the ensemble of isochromats:

$$\lim_{t \rightarrow \infty} \overline{M}_{xy}(\Delta\omega_0) = \int_{-\infty}^{\infty} \frac{M_{xy}^{\infty}(\Delta\omega)}{\pi} \frac{R_2'}{(\Delta\omega - \Delta\omega_0)^2 + R_2'^2} d\Delta\omega. \quad (41)$$

with the steady state transverse magnetization $\vec{M}^\infty = -\mathbf{A}^{-1}\vec{C}$ written out explicitly in equation (13):

$$M_{xy}^\infty = -\frac{M_0 R_1 \omega_1}{R_2 \omega_1^2 + R_1 (\Delta\omega^2 + R_2^2)} (\Delta\omega + iR_2) \quad (42)$$

At large t , the additional transverse decay characterized by T_2' enters only through Cauchy weighting in equation (41). As a consequence, T_2' effectively shifts the ensemble steady state up or down by changing how strongly off-resonant isochromats contribute to the average.

Behavior in the absence of T_1 and T_2 relaxation The T_2' behavior for large t in the limiting case of $T_1 \rightarrow \infty$ and $T_2 \rightarrow \infty$ deviates strongly from the behavior expected for transverse T_2^* relaxation in the absence of longitudinal relaxation. Without longitudinal decay, transverse T_2 relaxation must disappear for long time spans since no longitudinal magnetization remains to be converted into the transverse plane.

In the T_2' case, however, the transverse magnetization amplitude converges to a limiting value arising from the balance between continuous constant RF driving and static frequency dispersion. This follows naturally from the analytical solution obtained in subsection 3.1: The transverse magnetization is given by $M_{xy} = M_x + iM_y$. With the values computed in equation (36) for the case of $R_1 = R_2 = 0$ and $\vec{M}_0 \parallel \hat{z}$, this yields

$$M_{xy}(\Delta\omega, t) = M_0 \frac{\omega_1}{\Omega^2} [\Delta\omega - \Delta\omega \cos(\Omega t) + i \Omega \sin(\Omega t)]. \quad (43)$$

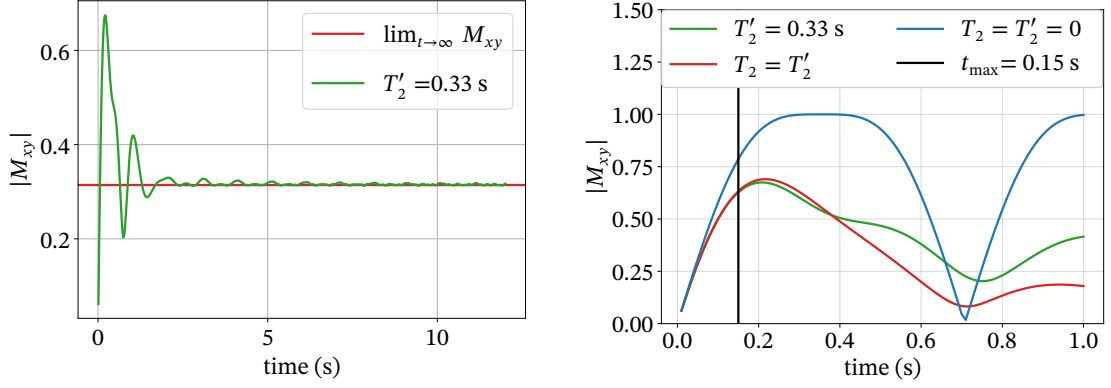
To compute the ensemble average for $t \rightarrow \infty$ of a Cauchy-distributed ensemble around the off-resonance offset $\Delta\omega_0$ with the homogeneous frequency spread R_2' , the magnetization is inserted into equation (14):

$$\lim_{t \rightarrow \infty} \overline{M}_{xy}(\Delta\omega_0) = \frac{1}{\pi} \lim_{t \rightarrow \infty} \int_{-\infty}^{\infty} M_{xy}(t, \Delta\omega) \frac{R_2'}{(\Delta\omega - \Delta\omega_0)^2 + R_2'^2} d\Delta\omega. \quad (44)$$

The oscillating parts of the integral disappear for large times. The remaining constant part can be evaluated symbolically with Mathematica and provides the steady state solution

$$\lim_{t \rightarrow \infty} \overline{M}_{xy}(\Delta\omega_0) = \frac{M_0 \Delta\omega_0}{\Delta\omega_0^2 + (R_2' + \omega_1)^2}. \quad (45)$$

Figure 3a shows that the simulated trajectory calculated by numerically integrating the analytic solution for one offset $\Delta\omega$ indeed converges to this value indicated by the red line.



(a) **limit $t \rightarrow \infty$ of T_2' decay** for $\Delta\omega_0 = 2\pi \text{ rad s}^{-1}$, $\omega_1 = 2\pi \text{ rad s}^{-1}$ and $M_0 = (0, 0, 1)^\top$ calculated by numerically integrating the analytic solution for one offset $\Delta\omega$.

(b) **limit $t \rightarrow 0$ of T_2' decay** for $\Delta\omega_0 = 2\pi \text{ rad s}^{-1}$, $\omega_1 = 2\pi \text{ rad s}^{-1}$ and $M_0 = (0, 0, 1)^\top$; Comparison of the numerically integrated analytical solution for $R_1 = R_2 = 0$ with T_2' dephasing (green) to the numerical solution for one isochromat with $R_1 = 0$ and no dephasing for two cases: with transverse T_2 relaxation set to the same value as T_2' (T_2^* approximation, red) and no transverse relaxation (blue); $t_{\max}$ marks the time at which the deviation between the full simulation and the T_2^* approximation exceeds 1%.

Figure 3: T_2' decay during the pulse in the limit of large and small time scales.

5.2 Approximation through T_2^* decay for short time spans

Unlike the long-term analysis in subsection 5.1, the simpler, relaxation-free case is considered first. This approach allows for the isolation of the effect of static frequency dispersion on short-time dephasing and the assessment of the applicability of an effective transverse relaxation time T_2^* .

Behavior in the absence of T_1 and T_2 relaxation In free decay, it is common to approximate the T_2' dephasing as an exponential decay by including it in the transverse relaxation as T_2^* (see subsection 2.1).

In the off-resonant case considered here, it is therefore natural to ask whether the resulting dephasing can, at least over short time spans, be approximated in an analogous way.

Figure 3b demonstrates that, for sufficiently short evolution, this is indeed a good approximation. For the chosen parameters in figure 3b, the difference between the T_2^* -model (red curve) and the full simulation (green curve) remains below 1% until $t_{\max} = 0.45 T_2'$, while completely neglecting relaxation (blue curve) already leads to a significant deviation at this time. Generally, the approximation breaks down when t becomes comparable to T_2' .

This behavior is physically intuitive. At very short times, spins with different off-resonance frequencies have not yet accumulated large relative phases, so the ensemble remains largely coherent. In this

regime, the effect of the static frequency distribution manifests primarily as a slow, monotonic loss of transverse magnetization, which is well captured by an effective transverse decay constant T_2^* . Only when the evolution time becomes comparable to T_2' do significant phase differences build up across the ensemble, leading to non-exponential dephasing and a breakdown of the T_2^* approximation.

As figure 4 shows, the time window over which the approximation remains accurate depends on the parameters $\Delta\omega_0$, ω_1 , and R_2' . In particular, the maximum time $t_{\max}$ up to which the deviation remains below 1 % increases with $|\Delta\omega_0|$ and decreases with increasing ω_1 and R_2' . As the dynamics are symmetric with respect to the sign of ω_0 , the absolute value is given.

This is consistent with the short-time argument, as stronger RF drive and broader offset distributions given by R_2' enhance $\Delta\omega$ -dependent mode-mixing effects.

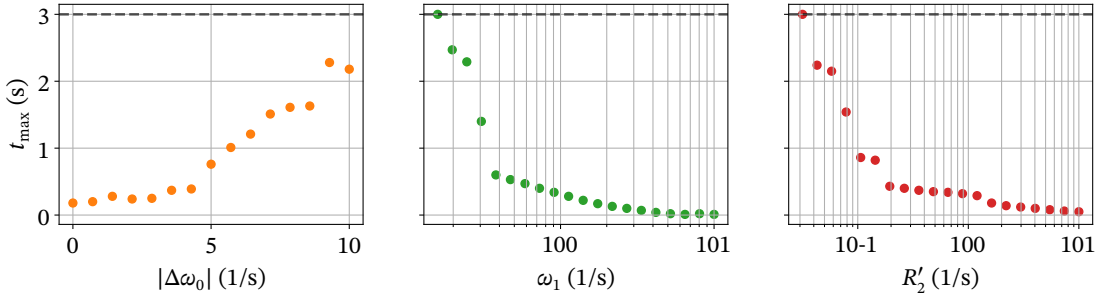


Figure 4: **Dependence of $t_{\max}$ on off-resonance, RF amplitude and R_2' :** The validity time $t_{\max}$ is defined as the largest time for which the deviation between the T_2' reference signal and the effective T_2^* model remains below 1%. The deviation is quantified by the RMS error between the magnitudes of the transverse magnetization signals, evaluated over a sliding time window to suppress oscillatory phase effects. Unless varied, parameters are fixed to $R_1 = 0$, $\Delta\omega_0 = 2\pi$ rad s⁻¹, $\omega_1 = 2\pi$ rad s⁻¹, and $R_2' = 1$ s⁻¹. The dashed line marks the end of the simulation. Both simulations were performed with the numerical approach, with $R_2 = R_2'$ and $R_2' = 0$ for the R_2^* model and with $R_2 = 0$ and $R_2' = R_2'$ for the reference.

Behavior in the presence of T_1 and T_2 relaxation In the presence of longitudinal and transverse relaxation, the short-term behavior is still dominated by the phase accumulation mechanism. At very small t , the relaxation terms R_1 and R_2 contribute only weakly, so that the transverse magnetization of each isochromat differs from the $R_1 = R_2 = 0$ case mainly by an additional slow, homogeneous attenuation. This additional attenuation must be considered in the calculation of T_2^* , but it does not change the applicability of the approximation.

6 Application in slice selection

In this section, the full T_2' simulation will be applied to slice selection, where the RF pulse, gradient-induced off-resonance, and relaxation act simultaneously. Numerically, this is realized by introducing another loop over resonance offsets into the Cayley-Klein simulation framework, chosen over

the other approaches based on the discussion in subsection 4.1. The source code is available in the accompanying public repository [16]. As the full simulation is too computationally expensive to be practical in day-to-day MR simulations, two approximation models will be tested: the T_2^* approximation and the hard-pulse approximation.

T_2^* approximation As discussed in subsection 5.2, ensemble-averaged dephasing may be approximated by an effective homogeneous transverse decay characterized by T_2^* as long as $t \ll T_2^*$. The exact validity range of this approximation depends on R_2' and the RF pulse properties.

Realistic slice select RF pulses involve substantially larger RF amplitudes than the $\omega_1 = 2\pi \text{ rad s}^{-1}$ considered in the preceding analysis. Since the deviation between the full T_2' simulation and the T_2^* approximation increases with increasing ω_1 and decreasing $\Delta\omega_0$ (see figure 4), the error is expected to be larger than in previous simulations and most pronounced in the slice center, where spins are closer to resonance and $\Delta\omega_0$ is smaller.

Hard-pulse approximation Another approximation commonly employed to reduce computational cost is the hard pulse approximation. Here, relaxation effects during the RF pulse are neglected entirely, and the effect of the RF pulse is treated as an instantaneous rotation taking place in the middle of the pulse, producing the same outcome as in the relaxation-free case. The phase is then rotated back to the middle of the pulse to avoid double-counting, and relaxation is subsequently applied during the free evolution in the remaining second half of the pulse.

While this approximation is advantageous due to its simplicity, it neglects any effect of the RF pulse on attenuation or phase dispersion accumulated during long RF pulses and is therefore expected to break down once the pulse duration becomes comparable to the relevant relaxation times.

Validity criterion The validity range of each approximation is assessed by comparing the magnetization profiles at the final time point with the full T_2' simulation.

Deviations are quantified for the magnitude of the transverse component and the longitudinal component. For $|M_{xy}|$, the maximum relative deviation is defined as

$$\varepsilon_{\max}^{xy} = \max_z \frac{| |M_{xy}^{\text{approx}}(z)| - |M_{xy}^{\text{full}}(z)| |}{\max_z |M_{xy}^{\text{full}}(z)|}. \quad (46)$$

with z denoting the position within the slice corresponding to the off-resonance offset. The normalization by the maximum transverse magnetization of the reference profile ensures that the error measure is insensitive to regions outside the excited slice.

For the longitudinal component, the magnetization $M_z(z)$ is compared directly, and the corresponding maximum relative error is defined as

$$\varepsilon_{\max}^z = \max_z \frac{|M_z^{\text{approx}}(z) - M_z^{\text{full}}(z)|}{\max_z |M_z^{\text{full}}(z)|}. \quad (47)$$

This worst-case error metric is chosen to detect localized breakdowns of the approximation models, which occur predominantly near the center of the slice, where the off-resonance is smallest.

An approximation is considered valid as long as the maximum relative error remains below 5%. This threshold corresponds to a level at which deviations in excitation amplitude or inversion efficiency remain small compared to other sources of error in practical MR experiments, such as B_1 inhomogeneity.

Simulation parameters In all simulations, T'_2 was set to the white matter value at 3 T in the human brain, 103 ms, calculated using equation (2) with $T_2 = 110$ ms reported by Wansapura et al. [17] and $T_2^* = 53.2$ ms taken from Peters et al. [18]. To avoid confusion, the intrinsic T_1 and T_2 are neglected. A realistic RF excitation is implemented as a Gaussian pulse with time-bandwidth product $R = 4$ and an apodization factor of 0.5, targeting a slice thickness of 5 mm, generated using the PyPulseq framework [19].

6.1 Results

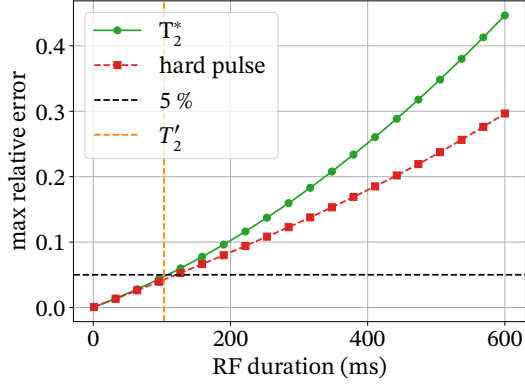
6.1.1 Valid timescale

Figure 5 shows the maximum relative error of the T_2^* approximation and the hard pulse approximation over the RF pulse duration for both 90° and 180° slice-selective pulses, as explained in section 6. The first column shows the error of the amplitude of the transverse magnetization $|M_{xy}|$, and the second column shows that of the longitudinal magnetization M_z , for a 90° excitation pulse in the first row and a 180° inversion pulse in the second, respectively.

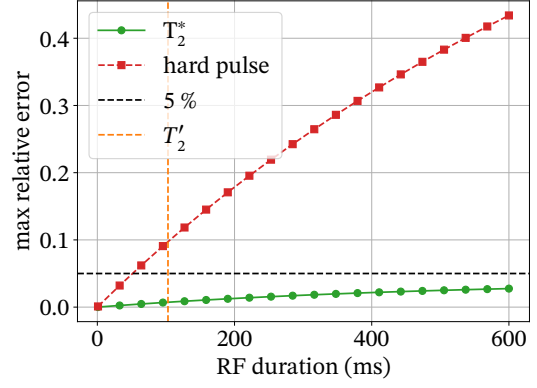
For RF pulse durations below 30 ms, the approximations show good agreement with the full T'_2 simulation for the longitudinal and transverse components at both flip angles. As the pulse duration increases, the error grows monotonically and exceeds the 5% threshold once the RF duration becomes comparable to the static dephasing time T'_2 , as indicated by the vertical dashed line. The exact time for which the threshold is exceeded depends on the flip angle and the component being considered. Generally, the approximations remain valid over longer pulse durations for the smaller flip angle.

Transverse magnetization magnitude For the transverse magnetization magnitude $|M_{xy}|$, both approximations exhibit similar behavior for 90° and 180° pulses. The error increases steadily with the RF duration and becomes substantial once the pulse length approaches T'_2 . Compared to the T_2^* approximation, the hard-pulse approximation yields smaller errors in the 90° case and slightly larger ones in the 180° case.

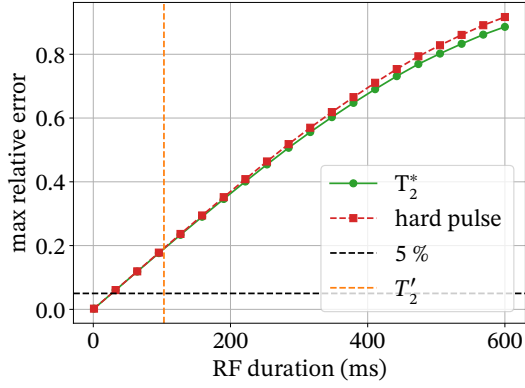
This behavior can be understood by examining how transverse relaxation is applied during the pulse. In the T_2^* approximation, relaxation acts continuously but only on the magnetization component that is currently in the transverse plane. In contrast, the hard-pulse approximation assumes an instantaneous rotation into the transverse plane and subsequently applies transverse relaxation to the full



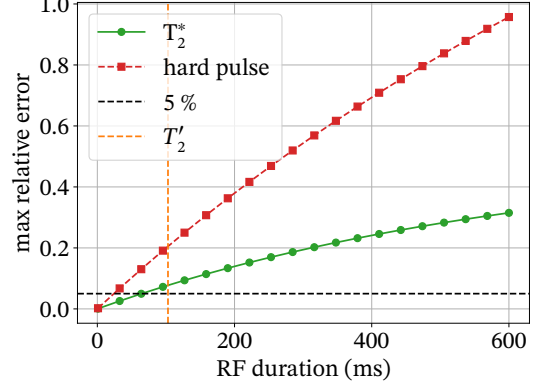
(a) Maximum relative error in $|M_{xy}|$ for a 90° excitation pulse



(b) Maximum relative error in M_z for a 90° excitation pulse



(c) Maximum relative error in $|M_{xy}|$ for a 180° inversion pulse



(d) Maximum relative error in M_z for a 180° inversion pulse

Figure 5: **Maximum relative deviation** of the approximation from the full T_2' simulation with $T_2' = 103$ ms. The RF pulse is given by a Gaussian PyPulseq event with time-bandwidth product $R = 4$ and apodization factor 0.5 to select a 5 mm slice. R_1 and R_2 are set to zero for clarity.

magnetization that ultimately contributes to $|M_{xy}|$, albeit only over half of the pulse duration. As a result, the two approximations differ mainly in the effective amount of transverse magnetization exposed to relaxation.

For the 90° pulse, the full T_2' simulation, unlike either approximation, exhibits a stronger attenuation in the slice center. This is then the maximum error, meaning that the approximation predicting a smaller effective transverse magnetization during the pulse yields closer agreement. Which approximation fulfills this condition depends on the RF pulse shape. For the Gaussian pulse considered here, the hard-pulse approximation leads to a larger effective relaxation-induced attenuation, and therefore, better agreement with the reference simulation.

Overall, this indicates that the timing of relaxation—whether applied continuously during the pulse or effectively after excitation—primarily affects the global attenuation of M_{xy} , while the detailed

slice-profile shape remains largely unaffected.

Longitudinal magnetization As neither approximation considers the steady-state-like long-term behavior of T'_2 dephasing comparable to a combination of transverse decay and longitudinal recovery for long RF pulses, the longitudinal slice profile exhibits higher errors than the transverse one. Still, the longitudinal magnetization M_z shows a different sensitivity to the approximation models. For both flip angles, the hard-pulse approximation exhibits a rapid increase in error with increasing RF duration, reflecting the complete neglect of the fact that magnetization is transverse and, therefore, subject to transverse relaxation for a finite time during the rotation. The T_2^* approximation, by comparison, remains accurate over a substantially wider range of pulse durations, particularly for the 90° excitation pulse, where the error in M_z stays well below the 5% threshold even for RF durations exceeding T'_2 .

Overall, these results confirm that the validity of both approximation models is governed primarily by the ratio between RF pulse duration and T'_2 , with the T_2^* approximation providing a more reliable description of longitudinal magnetization dynamics.

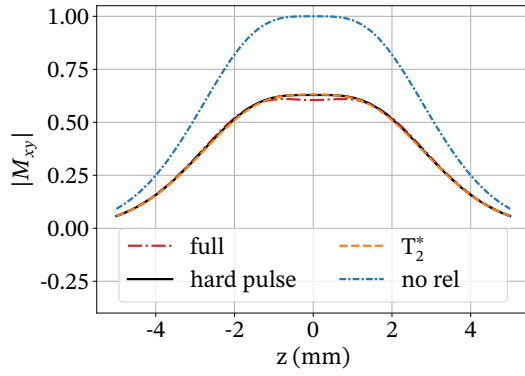
6.1.2 Excitation slice profile

To illustrate the mechanism underlying the validity limits defined above, figure 6 shows the slice profile and on-resonant magnetization trajectories for a representative 90° pulse with a pulse duration close to T'_2 , $T_{\text{pulse}} = 95.6$ ms.

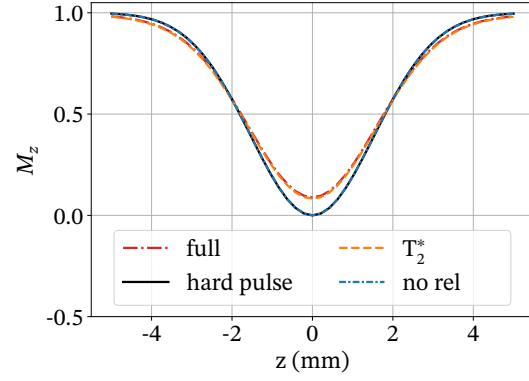
Figure 6a and figure 6b display the transverse magnetization magnitude $|M_{xy}|$ and the longitudinal magnetization M_z across the slice at the end of the RF pulse, respectively. The full T'_2 simulation is compared to the T_2^* approximation, the hard-pulse approximation, and a relaxation-free reference.

Transverse magnetization magnitude For transverse magnetization, both approximation models reproduce the overall slice profile reasonably well, with deviations primarily manifesting in the slice center, where the off-resonance resulting in the T'_2 decay is stronger than the off-resonance introduced by slice selection. The differences between the T_2^* and hard-pulse approximations remain small in $|M_{xy}|$, consistent with the validity analysis in Figure 5.

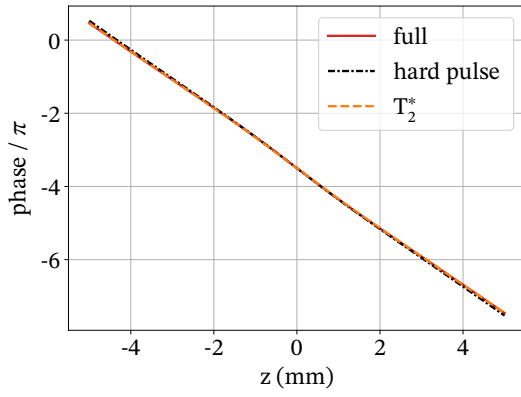
Phase dispersion across the slice The corresponding phase profiles shown in figure 6c exhibit an approximately linear phase spread, as no phase rewinder gradient was applied. The difference from the full simulation shown in figure 6d reveals that both simulations are accurate in the on-resonant case. However, for large flip-angles, the phase spread is not completely linear across the slice, and the deviations from linearity differ between the models. While the T_2^* approximation closely follows the full simulation, the hard pulse approximation yields stronger off-resonant errors. This confirms that replacing ensemble dephasing with an effective homogeneous decay captures the dominant phase dispersion during excitation, while the free-evolution phase trajectory of the hard-pulse approximation during the second half of the pulse causes significant errors in the off-resonant case.



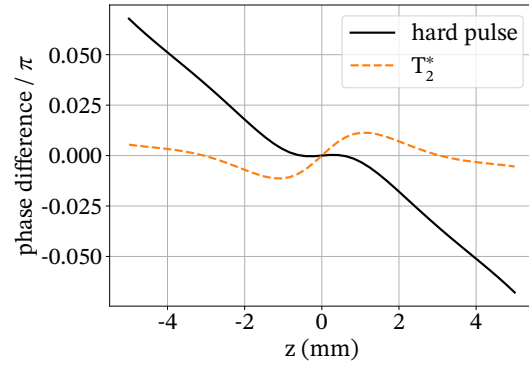
(a) Slice profile in $|M_{xy}|$



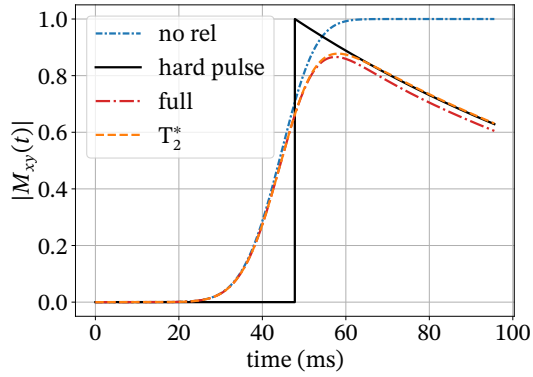
(b) Slice profile in M_z



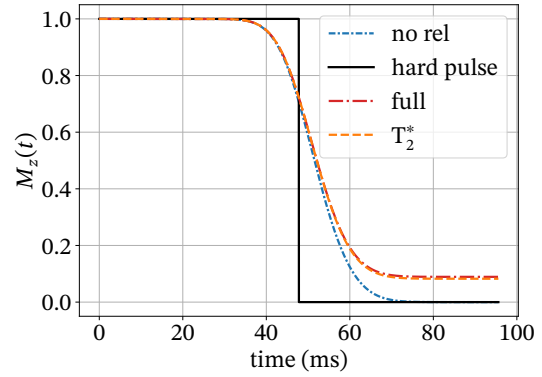
(c) Slice profile in the phase



(d) Slice profile in the phase difference to the full simulation



(e) Trajectory in $|M_{xy}|$



(f) Trajectory in M_z

Figure 6: **Maximum relative deviation** of the approximation from the full T_2' simulation with $T_2' = 103$ ms. The RF pulse is given by a gaussian pypulseq pulse with $R = 4$ to select a 5 mm slice. R_1 and R_2 were set to zero and the pulse duration was 95.6 ms

On-resonant magnetization trajectories Panels (e) and (f) of figure 6 show the on-resonant magnetization trajectories during the RF pulse, as this is where the deviation is most significant. The trajectory throughout the pulse in the hard-pulse approximation deviates strongly from the pulse, as expected, but produces essentially the same slice profile in $|M_{xy}|$. In M_z , the hard pulse approximation produces the slice profile of the relaxation-free case when R_1 and R_2 are neglected, leading to a large error. In contrast, the T_2^* approximation tracks the full simulation more closely, in agreement with the significantly smaller errors observed for M_z in the validity analysis.

Implications for the validity of the approximations Overall, this example demonstrates that while both approximations may yield comparable excitation profiles for short pulses, their underlying magnetization dynamics differ substantially once relaxation during the RF pulse becomes relevant. These differences are most apparent near the slice center and in the longitudinal component, explaining the error behavior observed in Figure 5.

6.1.3 Inversion slice profile

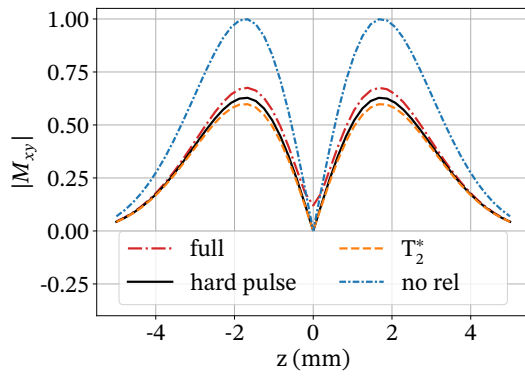
The corresponding example for a 180° inversion pulse is shown in figure 7. As before, the RF pulse duration is chosen close to T_2' , such that relaxation effects during the pulse are expected to play a significant role. Panels (a) and (b) show the transverse magnetization magnitude $|M_{xy}|$ and the longitudinal magnetization M_z across the slice at the end of the RF pulse, respectively.

Transverse magnetization magnitude For the transverse magnetization, both approximation models reproduce the qualitative structure of the slice profile, featuring a double-lobed shape caused by the inversion pulse. Compared to the excitation case, the deviations from the full simulation are more pronounced. The reduction in $|M_{xy}|$ relative to the relaxation-free case is stronger, reflecting the increased time that the magnetization spends near the transverse plane during inversion.

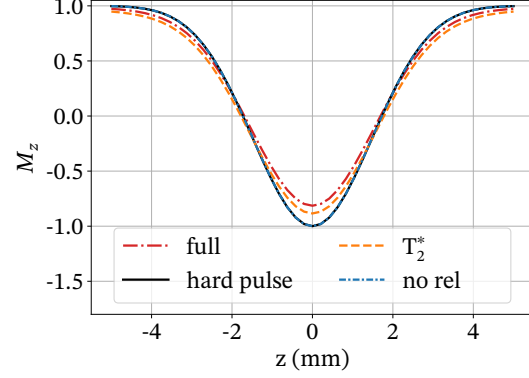
Phase dispersion across the slice The phase profile is shown in panel (c) with the difference to the full simulation in panel (d). Again, the phase difference across the slice is larger for the hard-pulse approximation. It is important to note that the unwrapped phase of 2π corresponds to 0, meaning that the phase difference is small in the off-resonant case and most significant in the center of the slice. This indicates that, for inversion pulses, replacing ensemble dephasing with a homogeneous decay captures only part of the accumulated phase dispersion during excitation.

On-resonant magnetization trajectories Panels (d) and (e) show the on-resonant magnetization trajectories during the RF pulse. For $|M_{xy}(t)|$, the hard pulse trajectory again deviates significantly from the full simulation, as is to be expected. At the end of the pulse, the T_2' dephasing causes a sign change absorbed into the phase jump also observed in panel (c), whereas the approximations simply return the transverse magnetization to 0, resulting in a large deviation.

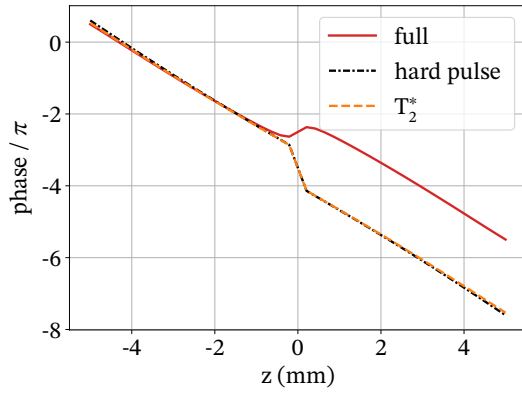
For the longitudinal component $M_z(t)$, the hard-pulse approximation exhibits an abrupt sign change



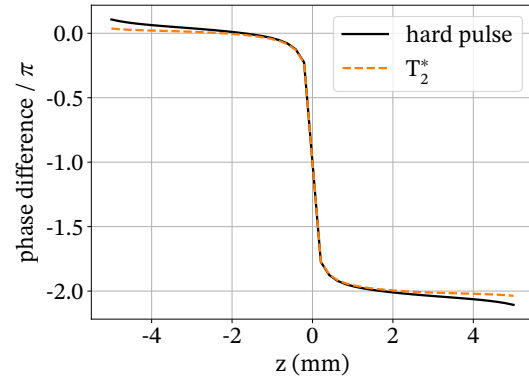
(a) Slice profile in $|M_{xy}|$



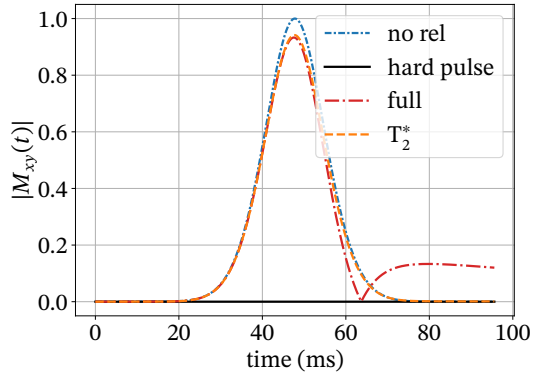
(b) Slice profile in M_z



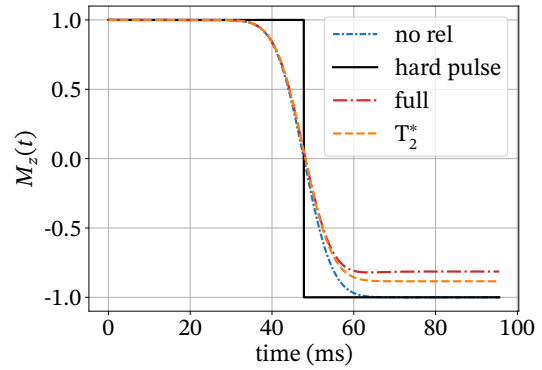
(c) Slice profile in the phase



(d) Slice profile in the phase difference to the full simulation



(e) Trajectory in $|M_{xy}|$



(f) Trajectory in M_z

Figure 7: **Maximum relative deviation** of the approximation from the full T_2' simulation with $T_2' = 103$ ms. The RF pulse is given by a gaussian pypulseq pulse with $R = 4$ to select a 5 mm slice. R_1 and R_2 were set to zero and the pulse duration was 95.6 ms

at the pulse center, while the full simulation and the T_2^* approximation show a smoother inversion accompanied by a reduced inversion efficiency. While the T_2^* approximation can partly capture this reduced efficiency, the hard pulse approximation shows strong deviations from the full simulation.

Implications for the validity of the approximations These differences highlight the increased sensitivity of inversion pulses to relaxation effects during the RF pulse and explain the earlier breakdown of both approximations observed in the validity analysis. Overall, the inversion slice profile demonstrates that relaxation during excitation plays a more critical role for 180° pulses than for 90° excitation. As a result, deviations between the approximation models and the full T_2' simulation become larger and appear at shorter RF pulse durations, consistent with the trends identified in Figure 5.

6.2 Discussion and practical implications

Overall, the hard-pulse approximation provides a reasonable description of slice-selective excitation when only the transverse magnetization magnitude $|M_{xy}|$ is of interest. For 90° pulses, the resulting excitation profiles remain close to those obtained from the full T_2' simulation as long as the RF pulse duration is short compared to T_2' . The approximation breaks down at shorter pulse durations for inversion pulses, as the dynamics are more complicated in this case.

Additionally, the hard-pulse approximation fails to accurately reproduce the phase of the transverse magnetization and the longitudinal component M_z . These quantities are strongly affected by the different application of relaxation throughout an off-resonant pulse, which is neglected in the hard-pulse model. As a result, significant phase errors and deviations in M_z arise even in regimes where the magnitude of M_{xy} appears well approximated.

In those cases, the T_2^* approximation shows better results, although it also breaks down for pulse durations close to T_2' . The deviations are generally strongest in the slice center while showing close agreement with the full simulation at the slice edges, where the off-resonance introduced by slice selection gradients dominates the off-resonance due to T_2' dephasing.

These results imply that the choice of approximation must be guided by the quantities of interest and the RF pulse duration relative to T_2' . For short pulse durations, the simple hard pulse approximation is the most practical. For applications employing longer pulses that focus on the excitation profile in the transverse plane, the hard pulse approximation is again the better choice, as it is more efficient and sometimes even yields better results for longer pulses. However, it should not be used when the pulse duration becomes comparable to T_2' and accurate off-resonant phase information or longitudinal magnetization dynamics are required, such as applications involving phase-sensitive contrast, refocusing schemes, or inversion recovery.

In these cases, the T_2^* approximation is more reliable, as it captures a substantial part of the relaxation

effects during the pulse while remaining significantly less expensive than the full T'_2 simulation. Nevertheless, even the T_2^* approximation becomes unreliable once the RF pulse duration approaches the static dephasing time, increasingly so in the slice center.

For long, low-bandwidth RF pulses or strong inversion pulses, a full simulation that explicitly accounts for T'_2 during excitation is therefore required to accurately predict slice profiles and magnetization dynamics. However, this is hard to implement due to the high computational cost of a full 3-dimensional simulation.

7 Conclusion and Outlook

In this research project, the T'_2 dephasing during off-resonant RF pulses was systematically investigated. To this end, several solution approaches to the off-resonant Bloch equations were developed and compared to each other, including direct numerical computation and an approximated analytical single-isochromat solution that was then ensemble averaged via Gauss-Legendre quadrature. Both methods were surpassed in computational efficiency by the Cayley-Klein-based simulation accelerated with numba.

This simulation was then used to assess common approximations in practical MR simulations. Slice selection was chosen as a simple example, but the obtained results are also applicable to other situations with external off-resonances adding to the ensemble-averaged off-resonances, such as CEST sweeps. The results show that the hard pulse approximation provides good results when mainly the transverse magnitude is of interest. In applications where the phase spread across the slice and the longitudinal component become important, the T_2^* approximation is the better choice. Both approximations are valid only as long as the pulse duration remains short compared to T'_2 and yield better results for stronger off-resonances $\Delta\omega_0$, as the off-resonance around which the Cauchy-distribution is centered dominates the Cauchy-distributed off-resonances, producing the T'_2 decay in this regime.

For longer pulses and the transverse component of inversion pulses, only the full simulation captures the relevant dynamics. However, while the simulation is feasible for the 1-dimensional slice, the computational cost increases significantly for a full 3-dimensional computation, which makes this unrealistic.

There are several ways to extend this analysis in order to improve the practical applicability of the reduced models.

First, the $|M_{xy}|$ error in the slice center, where deviations are most pronounced, increases approximately linearly with RF duration in the regime of interest and should depend on the RF amplitude and T'_2 . This suggests that a simple, parameter-dependent correction term may be feasible.

Second, the current hard-pulse approximation, which assumes only one instantaneous pulse followed by free evolution, completely neglects the transverse relaxation occurring during a 180° pulse. This might be improved by splitting the pulse into two 90° hard pulses or even finer intervals. Third,

a systematic mapping of regimes of dominance between slice-selection off-resonance and ensemble-averaged dephasing could provide further insights. Finally, the off-resonances stemming from microscopic field inhomogeneities were considered to be Cauchy-distributed in this project. While this is a common assumption yielding the well-known exponential decay in the free-evolution case, other distributions might indeed be closer to reality.

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A Appendix

A.1 Eigenvalue calculation

To obtain the eigenvalues of $\mathbf{A}$, the characteristic equation $|\mathbf{A} - \lambda \mathbf{1}| = 0$ is considered, which gives the cubic polynomial in λ ,

$$\lambda^3 + (R_1 + 2R_2)\lambda^2 + (R_2^2 + 2R_1R_2 + \Delta^2 + \omega_1^2)\lambda + (R_1R_2^2 + R_1\Delta^2 + R_2\omega_1^2) = 0. \quad (48)$$

In the off-resonance case, it is convenient to choose roots of the form $\lambda_1 = -R_{1\rho}$ and $\lambda_{2,3} = -R_{2\rho} \pm i\Omega$, where $R_{1\rho}$ is the slow relaxation rate in the effective field, $R_{2\rho}$ is an effective transverse damping term, and Ω is the precession frequency around the effective field. With this ansatz, the characteristic polynomial factorizes as

$$(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = (\lambda + R_{1\rho})[(\lambda + R_{2\rho})^2 + \Omega^2] = \lambda^3 + (2R_{2\rho} + R_{1\rho})\lambda^2 + (2R_{1\rho}R_{2\rho} + \Omega^2 + R_{2\rho}^2)\lambda + \Omega^2R_{1\rho} + R_{1\rho}R_{2\rho}^2 = 0. \quad (49)$$

Comparing coefficients with equation (48) gives the system

$$R_{1\rho} + 2R_{2\rho} = R_1 + 2R_2, \quad (50)$$

$$R_{2\rho}^2 + 2R_{2\rho}R_{1\rho} + \Omega^2 = R_2^2 + 2R_1R_2 + \Delta\omega^2 + \omega_1^2, \quad (51)$$

$$R_{1\rho}(R_{2\rho}^2 + \Omega^2) = R_1R_2^2 + R_1\Delta\omega^2 + R_2\omega_1^2. \quad (52)$$

This is now rewritten using the effective field angle θ . Substituting into Eqs. (51) and (52) and assuming $\Omega > R_1, R_2$ leads to

$$R_{1\rho}(\Delta\omega, \omega_1) = R_1 \cos^2 \theta + R_2 \sin^2 \theta, \quad (53)$$

$$R_{2\rho}(\Delta\omega, \omega_1) = R_2 \cos^2 \theta + R_1 \sin^2 \theta. \quad (54)$$

This gives the eigenvalues

$$\lambda_1 = -(R_1 \cos^2 \theta + R_2 \sin^2 \theta), \quad \lambda_{2,3} = -(R_2 \cos^2 \theta + R_1 \sin^2 \theta) \pm i\Omega. \quad (55)$$

A.2 Cofactor Method

Several derivations in this work make use of cofactors, both for constructing eigenvectors of singular matrices and for forming the inverse of a nonsingular matrix. For completeness, the general method is summarized here based on Horn and Johnson [20].

Let $\mathbf{B} \in \mathbb{R}^{3 \times 3}$ and let M_{ij} denote the 2×2 minor obtained by deleting row i and column j . The

corresponding cofactor is

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

Cofactor vector for singular matrices If $\mathbf{B}$ is singular, i.e. $\det \mathbf{B} = 0$, all its rows are linearly dependent. In this case, the cofactor vector formed from any row lies in the nullspace of $\mathbf{B}$:

$$\vec{v} = \begin{pmatrix} C_{i1} \\ C_{i2} \\ C_{i3} \end{pmatrix} \text{ satisfies } \mathbf{B} \vec{v} = \vec{0}.$$

This property is used in Sec. A.3 to construct eigenvectors of $\mathbf{A} - \lambda \mathbf{1}$, whose determinant vanishes at an eigenvalue.

Adjugate and matrix inverse For a nonsingular matrix $\mathbf{D}$, the cofactors assemble into the adjugate matrix,

$$\text{adj}(\mathbf{D}) = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix},$$

which satisfies the classical identity

$$\mathbf{D} \text{adj}(\mathbf{D}) = (\det \mathbf{D}) \mathbf{1}.$$

Thus the inverse is given by

$$\mathbf{D}^{-1} = \frac{1}{\det \mathbf{D}} \text{adj}(\mathbf{D}).$$

This cofactor-based formulation is used in the main text to obtain the form of $\mathbf{D}^{-1}$ once a simplified eigenvector matrix is chosen.

A.3 Eigenvector calculation

The eigenvectors are obtained as the solutions of

$$(\mathbf{A} - \lambda_k \mathbf{1}) \vec{v}_k = \vec{0}, \quad k = 1, 2, 3, \quad (56)$$

where λ_k are the eigenvalues calculated in subsection A.1. They can be calculated by constructing the matrix $\mathbf{B} = \mathbf{A} - \lambda$ using the cofactor approach explained in subsection A.2. Choosing the third row, the general eigenvector is obtained as

$$\vec{v}_k = \begin{pmatrix} \Delta\omega\omega_1 \\ \omega_1(R_2 + \lambda_k) \\ (R_2 + \lambda_k)^2 + \Delta\omega^2 \end{pmatrix} \quad (57)$$

With this, the eigenvectors are

$$\vec{v}_1 = \begin{pmatrix} \Delta\omega \omega_1 \\ \omega_1 \cos^2 \theta (R_2 - R_1) \\ \sin^4 \theta (R_1 - R_2)^2 + \Delta\omega^2 \end{pmatrix} \quad (58)$$

$$\vec{v}_2 = \begin{pmatrix} \Delta\omega \omega_1 \\ \omega_1 (\sin^2 \theta (R_2 - R_1) + i\Omega) \\ (\sin^2 \theta (R_2 - R_1) + i\Omega)^2 + \Delta\omega^2 \end{pmatrix} \quad (59)$$

$$\vec{v}_3 = \begin{pmatrix} \Delta\omega \omega_1 \\ \omega_1 (\sin^2 \theta (R_2 - R_1) - i\Omega) \\ (\sin^2 \theta (R_2 - R_1) - i\Omega)^2 + \Delta\omega^2 \end{pmatrix} \quad (60)$$